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Sensor Deployment Optimization Based on Optimal Recovery Interpolation

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ABSTRACT

Obtaining the desired signals in wireless sensor networks can be challenging due to various constraints on sensor placement or deployment. Retrieving the information accurately from sensors placed at non-uniform locations, is a problem of sensor communication and signal interpolation. In this research, the optimal recovery (OR) method, a deterministic framework that can use *a priori* bandwidth or spectral shape information, is used to interpolate from the given non-uniformly spaced samples. The error to be minimized is the maximum possible norm difference over a set of feasible signals. In the OR problem formulation and solution, the role of worst case feasible signals can be recognized but these signals are very difficult to find analytically. Computer simulations of feasible signals can help to produce estimates of the theoretical minimal worst-case error bounds. In this paper, monitoring of OR error bounds serves to assess sensor deployment configuration quality and to optimize the placement of additional sensors. Starting from an initial configuration of sensors, optimal deployment of additional sensors is clearly a more powerful option than random deployment of such sensors. These two approaches are compared and contrasted to show the improvement that is possible using the OR framework for one-dimensional and two-dimensional signals.

Keywords: sensor deployment, optimal recovery, min-max error, signal interpolation, feasible signals, wireless sensor networks.

1. INTRODUCTION

In wireless sensor network data collection arrangements, samples of fields of interest are obtained on irregular sampling grids in space and time [1]. The problem can be addressed using a wide variety of methods ranging from spline interpolation schemes to wavelet/multi-resolution signal representations on irregular grids [2]. In this paper, we continue our previous work [3-4] and apply the deterministic framework of Optimal Recovery (OR) [5-7] that can use *a priori* bandwidth or spectral shape information in the interpolation. In addition, analysis of worst case errors and the signals which cause these errors can be used to assess the quality of a configuration of sensors and to optimize the placement of additional sensors. An earlier version of this work can be found in [8].

The formulation that makes the recovery approach optimal is a min-max error criterion that uses information about the values and location of the samples as well as the knowledge of the signal's energy bound and its bandwidth [7]. The error to be minimized is the maximum possible over all the feasible signals. Thus, the error depends highly on the location of the available samples/sensors since this determines the signal(s) that would produce the worst case error at one fixed interpolated sample location (linear functional estimation) or on multiple interpolated samples (linear transformation estimation). In this paper, the computation of these error bounds is done numerically by simulation of the set of feasible signals.

Examples are presented using low-pass bandlimited 1-D test signals. More complex examples are included using a 2-D test signal with non-separable spectral support. Finding the locations of the worst case errors provides a strategy to optimally add new sensors to an existing configuration. The optimal addition of sensors is illustrated and compared with the approach of adding sensors in a random fashion.

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2. THEORY AND METHODOLOGY.

2.1 Theory of Optimal Recovery

In this section we briefly review the theory of optimal recovery as applied to the interpolation problem [7] including the reconstruction of signals from non-uniform samples. The theory of optimal recovery provides a broader setting for the problem of interpolation by casting it in the setting of signal spaces. The OR framework focuses on feasible signals and worst-case (extremal) signals which achieve the error bounds [5-7].

2.1.1 Linear Estimates for One-Dimensional Bandlimited Signals.

A a-bandlimited signal $f(t)$ can be represented as an orthogonal expansion using its equally spaced samples $f(kT_o)$, k=0,±1,±2,±3....., for the sampling period $T_o = \pi / a$. If the signal is sampled using $T = \pi / \sigma$ any $\sigma > a$, it is said to be oversampled and it is represented by the nonorthogonal interpolation formula

$$f(t) = \sum_{k=-\infty}^{\infty} f(kT) \frac{\sin a(t-kT)}{\sigma(t-kT)}$$

When $f(t)$ is approximated using N uniform samples $f(t_k)$, $t_k=kT$, k=1, 2,....N, the above equation could be simply truncated as follows

$$f_N(t) = \sum_{k=1}^{N} f(kT) \frac{\sin a(t-kT)}{\sigma(t-kT)} \ ,$$

however, this is not an optimal approximation of $f(t)$. The Optimal Recovery (OR) theory of deterministic estimation of a signal $f(t)$ from N linear measurements is applicable in the Hilbert space β_a of a-bandlimited signals. Consider the given measurements as an N-vector $\underline{y}$ with entries $y_k = f(t_k)$, k=1,2,3,...,N. Equivalently, the given samples are the values of an information linear transformation $\underline{y}=I(f)$ that produces the N samples from the signal $f(t)$ [6]. The signal can be estimated at any arbitrary location t_0 using the OR framework for the specific case of Bounded Linear Functional (BLF) estimation. Let us also assumed that the a-bandlimited signal has a known bound on its energy given by $\|f\|^2 = \int_{-\infty}^{\infty} |f(\tau)|^2 d\tau \le \varepsilon^2$ so that a feasible signal belongs to the set

$$R = \left\{ f \in \beta_a : f(t_k) = y_k;\ k = 1,2,3,\cdots,N; \|f\|^2 \le \varepsilon^2 \right\}.$$

Note that the null-space of the given information transformation I is

$$\aleph = \left\{ f \in \beta_a : f(t_k) = 0; k = 1,2,3,\cdots,N \right\}.$$

The optimal estimate and the worst case error to be minimized in the estimation of a BLF $f(t_0)$ are:

$$f_{t_0}^o = \underset{f_{t_0}}{\arg\min} \left\{ E_{t_0} \right\} \ ; \ E_{t_0} = \sup_{f \in R} \left| f(t_0) - f_{t_0} \right| .$$

A powerful result from this theory is that the optimal solution at any fixed but arbitrary t_0 is always the mid-point of the range of possible values of $f(t_o)$ and that this value is $f_{t_0}^o = f_{MN}(t_0)$ where

$$f_{MN}(t) = \sum_{k=1}^{N} w(k) \frac{\sin a(t-t_k)}{\pi(t-t_k)}$$

is the minimum norm (energy) signal in R. To find the weights $w(k)$, forming the vector $\underline{w}$ we simply enforce the interpolation property of any $f \in$R: $f_{MN}(t_m) = y_m$, m=1,2,....N. In matrix form this produces the solution:

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$$\underline{w} = S^{-1}\underline{y} \;\; ; \;\; [S]_{m,k} = \frac{\sin a(t_m - t_k)}{\pi(t_m - t_k)} \quad .$$

Once these weights are calculated from the known samples, $f_{MN}(t)$ can be evaluated at any location. The resulting minimum worst case error is thus

$$E^o_{t_0} = \sup_{f \in R} \left| f(t_0) - f_{MN}(t_0) \right|$$

A more general problem can be similarly formulated to estimate K values simultaneously rather than a fixed but arbitrary time sample. Let $U(f)$ be the set of K desired samples of $f(t)$ (a linear transformation from β_a to $\Re^K$), the criterion to minimize in this version of the OR problem is

$$\underline{z}^o = \arg\min_{\underline{z} \in \Re^K} \{E_U\} \; ; \; E_U = \sup_{f \in R} \left\| U(f) - \underline{z} \right\|_p$$

where p is the usual parameter in the ℓ_p norm for K-vectors.

$$\left\| \underline{z} \right\|_p = \left(\sum_{k=1}^{K} |z_k|^p \right)^{1/p} \quad p < \infty \;\; ; \;\; \left\| \underline{z} \right\|_\infty = \max_k \; \{ |z_k| \}$$

The solution is also obtained in terms of the minimum norm signal $f_{MN}(t)$ so that $\underline{z}^o = U(f_{MN})$ and the resulting minimum error is

$$E^o_U = \sup_{f \in R} \left\| U(f) - U(f_{MN}) \right\|_p \tag{1}$$

2.2 Extremal Signals, Null-Space Signals, and Error Bounds.

The Optimal Recovery problem for the estimation of a fixed but arbitrary sample (a bounded linear functional) is tractable using a Hilbert space β_a since the desired and the given time samples can be represented as inner products as follows:

$$f(t_0) = \langle f, \phi_0 \rangle \; ; \; \phi_0(t) = \frac{\sin a(t - t_0)}{\pi(t - t_0)} \;\; ; \; f(t_k) = \langle f, \phi_k \rangle \; ; \; \phi_k(t) = \frac{\sin a(t - t_k)}{\pi(t - t_k)} \; ; \; k = 1,2,\ldots,N$$

The error bound to be minimized and its optimal value are thus:

$$E_{t_0} = \sup_{f \in R} \left| \langle f, \phi_0 \rangle - f_{t_0} \right| \;\; ; \;\; E^o_{t_0} = \sup_{f \in R} \left| \langle f, \phi_0 \rangle - \langle f_{MN}, \phi_0 \rangle \right|$$

and from the Schwarz inequality $\left| \langle f, \phi_0 \rangle - \langle f_{MN}, \phi_0 \rangle \right| = \left| \langle f - f_{MN}, \phi_0 \rangle \right| \le \| f - f_{MN} \| \, \| \phi_0 \|$, however, this bound cannot be achieved for $f \in R$ with an arbitrary ϕ_0. Noting the role of the two subspaces

$$S_N = Span\{ \varphi_k \, ; k = 1,2,3, \cdots, N \} \;\; ; \;\; \aleph = S_N^\perp = \{ f \in \beta_a : f(t_k) = \langle f, \varphi_k \rangle = 0 \; ; k = 1,2,3,\cdots,N \}$$

we can see that any $f \in R$ can be decomposed as $f = f_{MN} + f^\perp$ and similarly $\phi_0 = \hat{\phi}_0 + \phi_0^\perp$ so that the largest error is achieved with the largest energy $f^\perp$ projected onto the direction of ϕ_0 leading to

$$E^o_{t_0} = \sup_{f \in R} \left| \langle f - f_{MN}, \hat{\phi}_0 + \phi_0^\perp \rangle \right| = \left| \langle f_e^\perp, \phi_0^\perp \rangle \right| = \| f_e^\perp \| \, \| \phi_0^\perp \| = \sqrt{\left(\varepsilon^2 - \| f_{MN} \|^2 \right) \| \phi_0 - \hat{\phi}_0 \|^2} \tag{2}$$

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where $f_e = f_{MN} + f_e^{\perp}$ is an *extremal signal*, a feasible signal with maximum energy ε^2 which produces the minimum worst-case error so that

$$E_{t_0}^{o} = \sup_{f \in R} \left| f(t_0) - f_{MN}(t_0) \right| = \left| f_e(t_0) - f_{MN}(t_0) \right| = \left| f_e^{\perp}(t_0) \right| \quad . \tag{3}$$

We also note the important role of the *extremal null signal* $f_e^{\perp} \in \aleph$ which produces the largest error. The more general extremal signals and the error bounds in Eq. (1) are more difficult to generate theoretically [5-6]. In the next subsection, we use computer simulations to produce error bounds and extremal signals for the problem of estimating $f(t)$ over a set of K uniform samples over a Region of Interest (ROI) in the form of a finite interval $[T_1,T_2]$.

2.3 Experimental Generation of Feasible Signals, Null-Space Signals and Error Bounds.

The numerical approach to generating feasible signals is simply to directly generate a very large number of bandlimited, bounded energy signals that interpolate a fixed set of samples and that are located in a finite interval. Here we first produce test signals which are sums of 10 *sinc* functions with random amplitudes and random time shifts. The amplitudes are chosen to take the form $A_i = (B_i)^{1/7}$ where the B_i are uniformly distributed random values, this way, the A_i are non-uniformly distributed and concentrated near the edges of the interval [-0.3953, 0.3953], see Fig. 1. Each sinc function is also subjected to a uniform random time shift in the range [0,40]. Thus, the test signals are of the form

$$g(t) = \sum_{i=1}^{10} A_i \frac{\sin \pi(t - r_i)}{\pi(t - r_i)}$$

so that they are all bandlimited to $a=\pi$ radians/second. These signals are used to produce feasible signals by selecting the first (reference) test signal $f_1(t)$, sampling it on a fixed chosen set of N=30 non-uniform locations t_k, k=1,2,…,N. The given samples are also within the ROI interval [0,40]. From these samples we obtain the reference minimum norm interpolating signal f_{MN}(t). Each test signal $g(t)$ is used to produce a *null-space signal* by subtracting it's own projection onto the subspace S_N which is $g_{MN}(t)$. The form of each feasible signal is therefore:

$$g_F(t) = f_{MN}(t) + \left(g(t) - g_{MN}(t)\right) \; ; \; \|g_F\|^2 \le \varepsilon^2 \quad . \tag{4}$$

The energy bound is used to discard generated feasible signals that have energy beyond the desired bound ε^2. In this subsection we use 5000 test signals whose energy distribution is shown in Fig. 1 and such that 4529 bounded energy feasible signals are chosen for the bound ε^2 =1. Figure 2 shows the reference test signal $f_1(t)$ and the reference minimum norm signal $f_{MN}(t)$ produced from it's samples at the shown non-uniform locations.

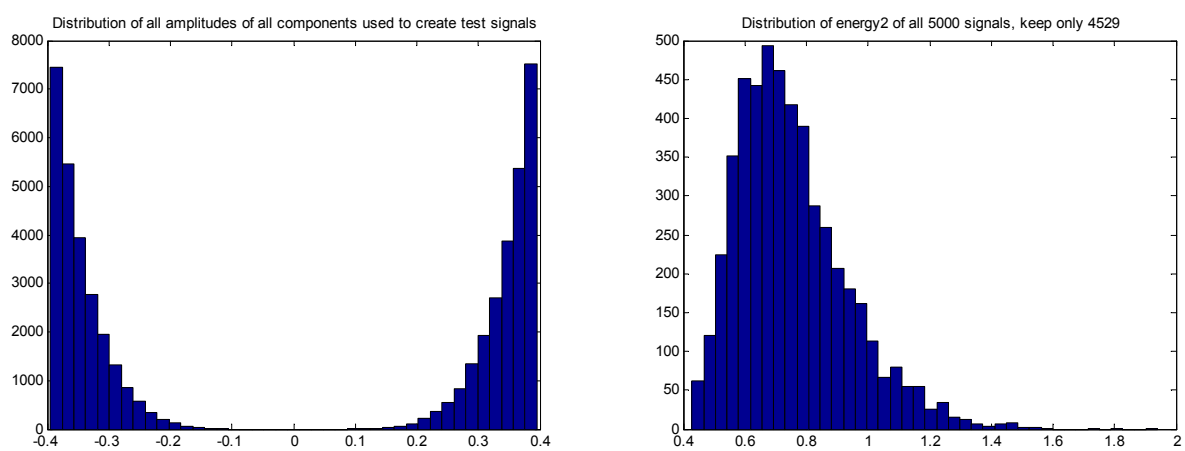

Fig. 1. Left: Distribution of 50,000 amplitudes used to produce 5,000 test signals. Right: Distribution of the energy of all 5000 test signals from which 4529 feasible signals are created which have energy bounded by 1.0.

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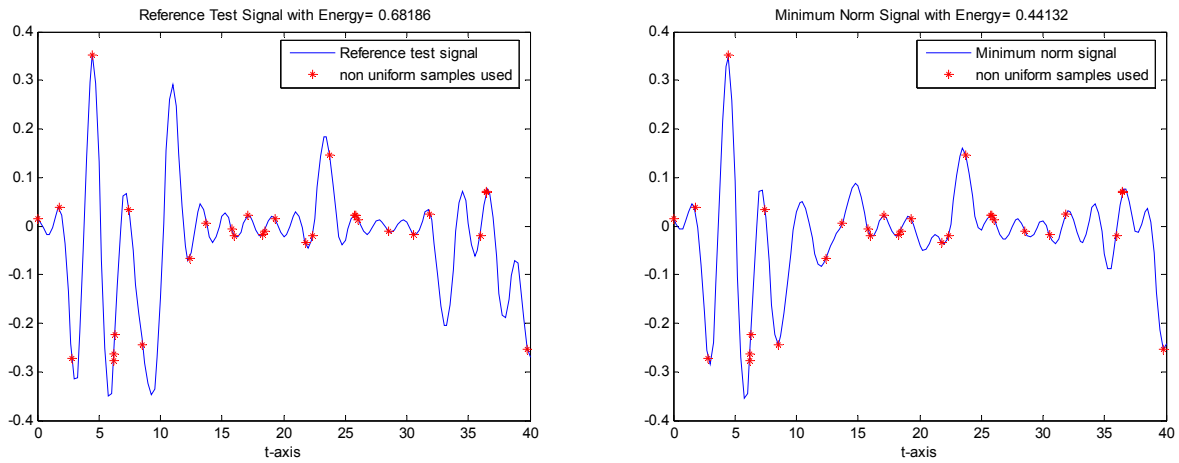

Fig. 2. Left: The first or reference test signal with energy 0.68186 which is used to produce the 30 samples shown. Right: The Minimum Norm signal that goes through these samples and which has energy 0.44132.

Figure 3 shows all of the feasible signals together with the reference minimum norm signal. From figure 3 we can estimate the worst case error bound corresponding to Eqs. (2) and (3) at each time location in the ROI. This bound is the maximum deviation of the feasible signal values from the reference minimum norm signal. This maximum deviation is the same as the maximum of all the amplitude of all the null-space signals $g(t)-g_{MN}(t)$ as noted in Eq. (3). The right hand side of Fig. 3 shows the error bound, also call the *range error*, which is one-half of the total range of the null signal amplitudes at each time location within the ROI.

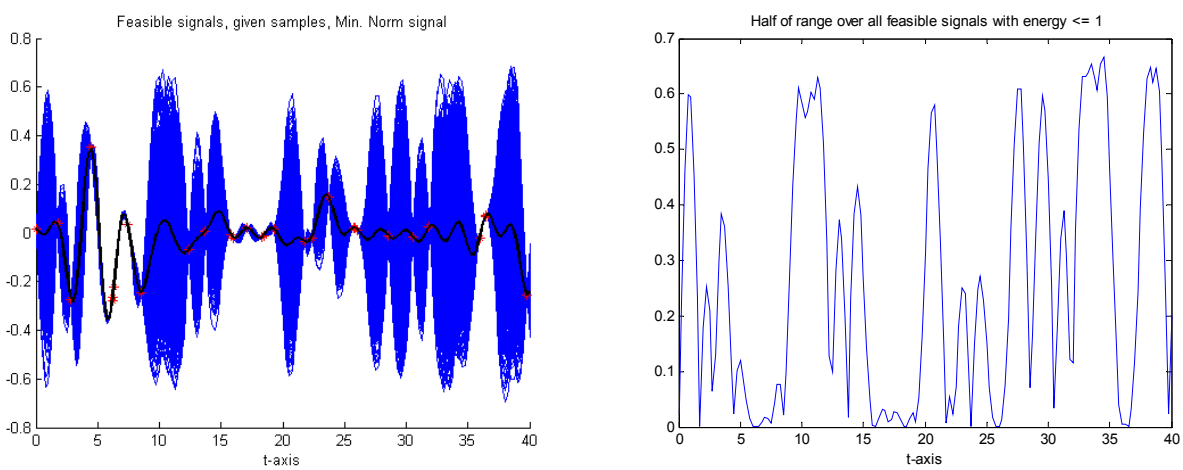

Fig. 3. Left: All bounded energy feasible signals which go through the same fixed samples together with the reference minimum norm signal. Right: The range error, one-half of the amplitude range of the feasible signals at each time sample location and which is the estimated minimal worst case error bound for arbitrary time sample estimation.

For the estimation of K uniform time samples simultaneously, Fig. 4 shows on the left side the distribution of the values of the three standard ℓ_p norms over the ROI [0,40] for all 4529 feasible signals. The maximum ℓ_p norm gives the experimental derivation of the error bound in Eq. (1) for the linear transformation U consisting of sampling $f(t)$ on a dense grid over the ROI using a sampling period of Δ_t=0.25 sec. to produce K=161 samples. Figure 4 also shows the three corresponding extremal signals for each of the three standard norms. These extremal signals achieve the largest deviation from the reference minimum norm signal over the ROI.

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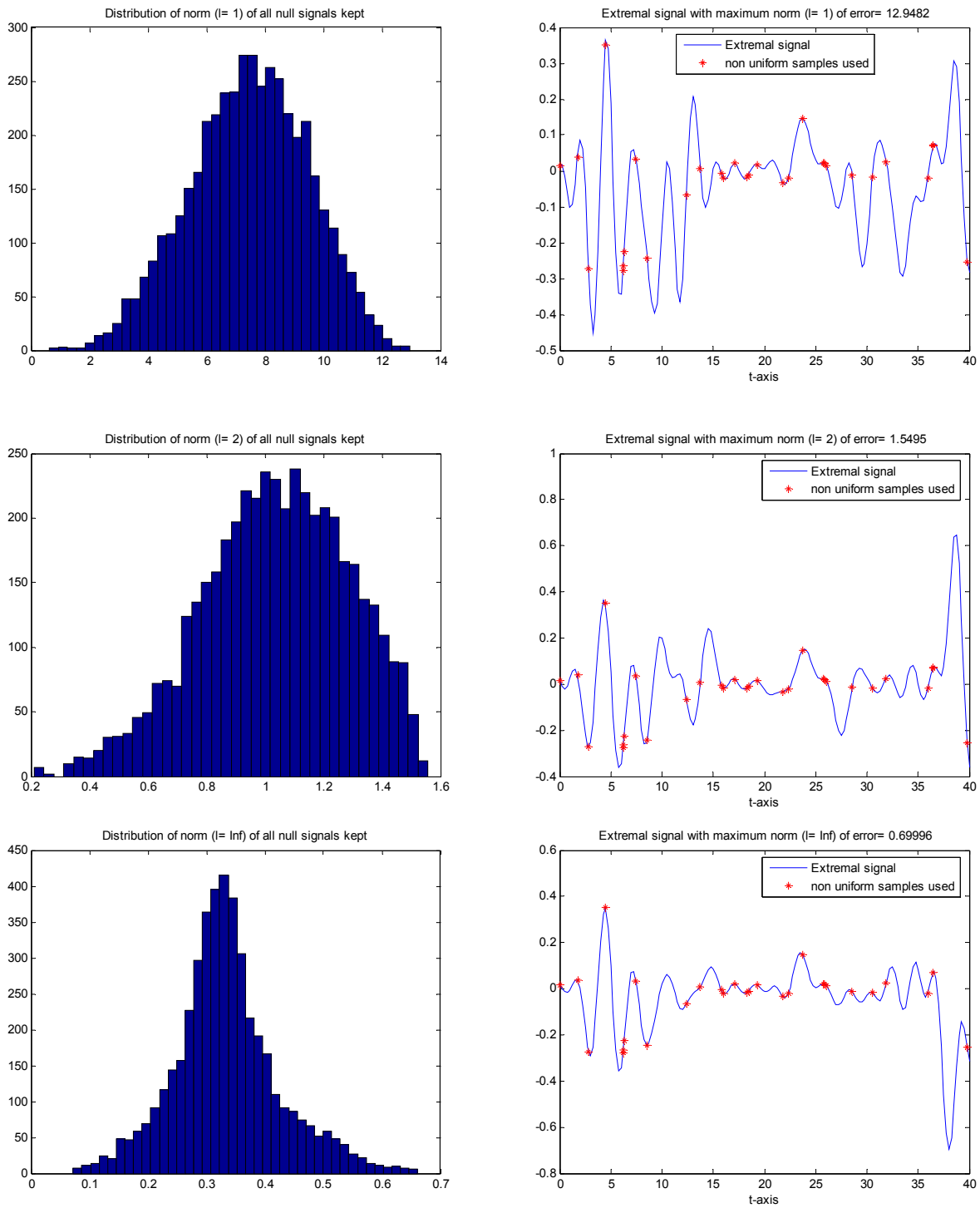

Fig. 4. Left: Distributions of the values of the three standard norms, ℓ_1, ℓ_2, and ℓ_∞, of all the null signals corresponding to all of the 4529 feasible signal used. Right: The extremal signals that produce the worst case error for each of the three standard norms.

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3. SENSOR DEPLOYMENT ANALYSIS USING OPTIMAL RECOVERY ERROR BOUNDS.

3.1 Problem Set-up for One-Dimensional Signals.

In this section we begin by illustrating a one-dimensional problem for the approach to optimal sensor deployment. The approach consists of placing new sensors at well chosen locations in order to decrease the OR error in Eqs. (1) or (2). Let us consider again the generation of test signals and feasible one-dimensional bandlimited signals for this application. We generate 600 test signals that are known to be bandlimited to π rad./sec which is less than an assumed bandwidth of a=1.5π rad./sec. Each test signal takes the form

$$f(t) = A_1 \frac{\sin \pi(t-r_1)}{\pi(t-r_1)} + A_2 \frac{\sin \pi(t-r_2)}{\pi(t-r_2)} + A_3 \frac{\sin \pi(t-r_3)}{\pi(t-r_3)} \tag{4}$$

where A_1, A_2, A_3 are random amplitudes in the range [-1.0,1.0] and r_1, r_2, r_3 are random shifts in the range [0,40]. The test signals produced by Eq. (4) are theoretically bounded by a large number $\varepsilon^2 = 9a/\pi = 13.5$, however, this limit is rarely achieved experimentally. Most importantly, the exact bound on the energy ε^2 can be arbitrary in the optimal sensor deployment scheme since only the relative reduction of the error is important. Again, the first test signal $f_1(t)$ is considered as a reference signal to determine a set of sample locations and signal values at these locations. The generation of the feasible signals follows the same methodology as in the previous section where the important roles of the feasible signals and the null-space signals are discussed.

For experimental purposes, all signals are evaluated in the time interval [0, 40] using a sampling period of T_s=0.1. In addition, the given samples will be constrained to be taken on two disjoint intervals which form two ROIs [5,15] and [25,35]. The additional sensors/samples are also to be added in these two ROIs. Ten non-uniformly placed samples are used within each ROI to reconstruct the reference minimum norm signal $f_{MN}(t)$ from the samples of the reference test signal $f_1(t)$. All the feasible signals pass through the given non-uniform samples, as shown in Fig.5. As mentioned earlier, Fig. 5 can be used to clarify the estimation of the minimal worst case error bounds by simply measuring half of the maximum range of feasible signal values. Note that samples at time values near the given samples, within the two ROIs, have smaller range error than those outside of the two ROIs. The error for sample locations far away from the given samples is larger but is bounded by a maximum value that can be derived from Eq. (2) with $f_{MN}(t_0) = 0$ and $\hat{\phi}_0 = 0$:

$$\sup\left\{E_{t_0}^{o}\right\} = \sup_{f \in R} \left|f(t_0)\right| = \varepsilon \left\|\phi_0\right\| = \varepsilon\sqrt{\frac{a}{\pi}}$$

since $f_{MN}(t_0)$ is approximately 0 at those locations and $\phi_0(t)$ is approximately orthogonal to all $\phi_k(t)$, k=1,2,…,N. We also note from the left side of Fig. 5 that the minimum norm signal goes through the mid-point of the range of values of all feasible signals. The plot of half of the difference between the maximum and minimum values of feasible signals at each location, which is the *range error*, is shown on the right side of Fig. 5. Again, it can be observed that the range error is smaller in the ROIs compared to other regions where samples are not present.

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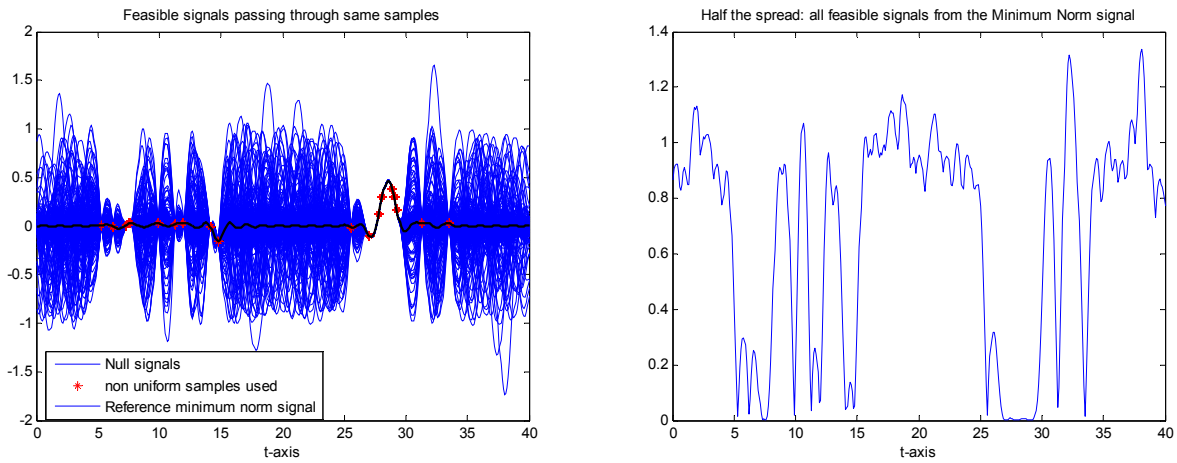

Fig. 5. Left: All feasible signals passing through the same samples in the two ROIs consisting of the intervals [5,15] and [25,35]. Right: The corresponding range error which is half the spread between the maximum and minimum values achieved

3.2 Optimal Deployment of Additional Sensors for 1-D Signals.

The approach to sensor deployment optimization is described next for this set-up. Initially, we place a sensor (sample) within each ROI at the location where the range error is largest in order to attempt to decrease the error by a maximum possible amount. Hence the range error will be zero at that location after this step. This process is repeated until the desired number of new additional sensors is reached. The plot of optimal deployment of sensors after adding 8 new sensors in each ROI is shown in Fig.6 (left). The new range error for this new configuration of sensor locations is shown in Fig.6 (right). The error in the regions of interest is clearly much lower throughout each ROI and is nearly equal to zero.

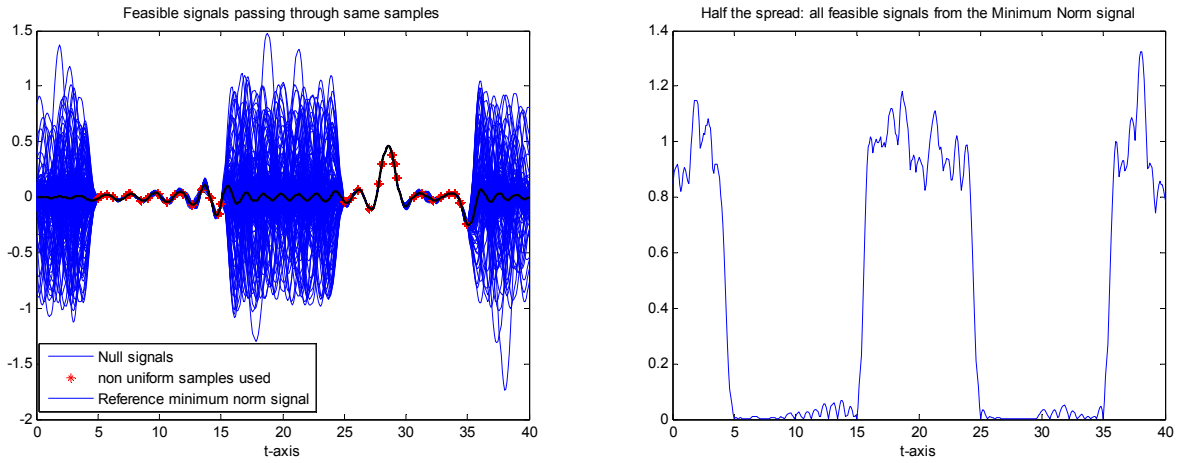

Fig. 6. Left: All feasible signals passing through the increased set of samples in each of the two ROIs after the addition of 8 new sensors in each of the two ROIs. Right: The corresponding range error after optimal sensor deployment.

3.3 Random Deployment of Additional Sensors for 1-D signals

Random deployment of sensors is mainly implemented in order to compare with the optimal deployment of sensors described in the previous subsection. As in the example showing optimal deployment, ten non-uniformly placed samples in each ROI are used initially to determine range error. In this case, one new, randomly-located sensor is introduced at every iteration in each ROI. This is repeated for eight iterations and compared to the optimal deployment

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case. The metrics used for comparison of both methods is the average of the range error function over each of the two ROIs. Figure 7 shows the plot of this metric for optimal deployment vs. random deployment for the two ROIs. The nine values of the metrics correspond to 0,1,2,…,8 additional sensors in each ROI.

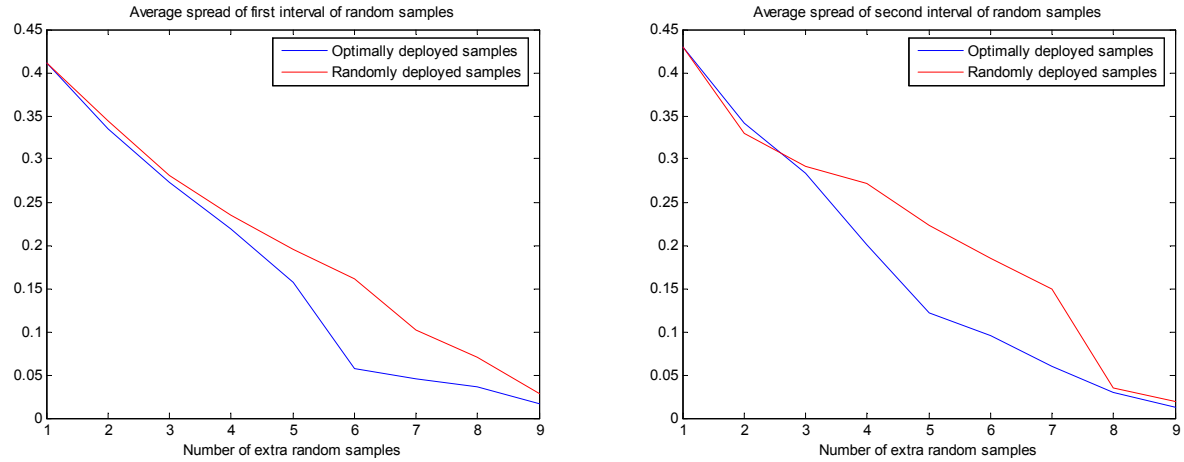

Fig. 7. Left: The average range error over the first ROI for optimally and randomly deployed additional sensors. Right: The average range error over the second ROI for optimally and randomly deployed additional sensors.

3.4 Problem Set-up for Two-Dimensional Signals.

Deployment of additional sensors is easily extended to two-dimensional signals with known spectral support. Let us consider 60 signals, each of which is a sum of five modulated Gaussian wavelets. For consistency with previous work [3-4], the two independent variables are denoted t and x corresponding to time and space respectively. Each test signal has the following form:

$$z(t,x)=\sum_{i=1}^{5} A_i\, e^{-\pi\left(\frac{x-s_i}{\xi_o}\right)^2} e^{-\pi\left(\frac{(t-r_i)-(P_o(x-s_i))}{\tau_o}\right)^2} \cos\left(2\pi f_o((t-r_i)-(P_o(x-s_i))+\theta_o\right)$$

where A_i's are the random amplitudes and the r_i's and s_i's represent the random shifts of the signal in the t-direction and x-direction respectively. The parameters used for all test signals are: ξ_o=4, τ_o=4x10^{-4}, θ_o=0, P_o=10^{-5}, and f_o=12,500 Hz. The reference test signal, from which the samples are taken, is shown in Fig. 8 (left) within a rectangular region [-3.2x10^{-4},-3.2 x10^{-4}] in the t-direction by [-5, 5] in the x-direction. This region surrounds the smaller Region of Interest (ROI) where all the sensors are located and which is the rectangle [-2x10^{-4},2x10^{-4}] in t-direction by [-2,2] in x-direction. The sampling periods chosen to deal with these signals are Δ_t =10^{-5} and Δ_x =0.125. Initially, N=99 non-uniform samples are available within the central ROI region. The signals are assumed to be bandlimited to half of the region within [-20,000,20,000] in the temporal-frequency direction and [-0.8,0.8] in the spatial-frequency direction, see the right hand side of Fig. 8. The optimal recovery method is applied to first find the reference minimum norm signal $z_{MN}(t,x)$ from the samples taken from the reference test signal $z_1(t,x)$. Following the 2-D version of Eq. (4), 60 feasible signals are obtained from the corresponding 60 null-space signals. The difference between the 60 test signals and their minimum norm versions are null-space signals which are added to $z_{MN}(t,x)$ to produce feasible signals (not shown). The worst case error bounds at arbitrary t-x sample locations can be obtained directly from the null-space signals as noted earlier for the 1-D case based on Eq. (3). The plot of the range error, the maximum amplitude of all the null-space signals at each sample location, is shown in Fig. 9 within and surrounding the ROI. The range error is clearly much smaller within the ROI since that is the region containing the N samples.

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3.5 Optimal Deployment of Additional Sensors for 2-D signals

Optimal deployment of sensors is performed in the same way as described in the 1-D case. Using these null-space signals, the maximum range error value and location is calculated over the entire ROI. An additional sensor is placed at this location which is the optimal sensor placement strategy. Clearly, the range error will be zero at that location after this step. This operation is repeated until we add 10 sensors to the existing configuration of 99 sensors. The goal of this approach is to have new sensors placed at maximum range error locations in order to attempt to maximally reduce the ROI-wide average range error at the next iteration.

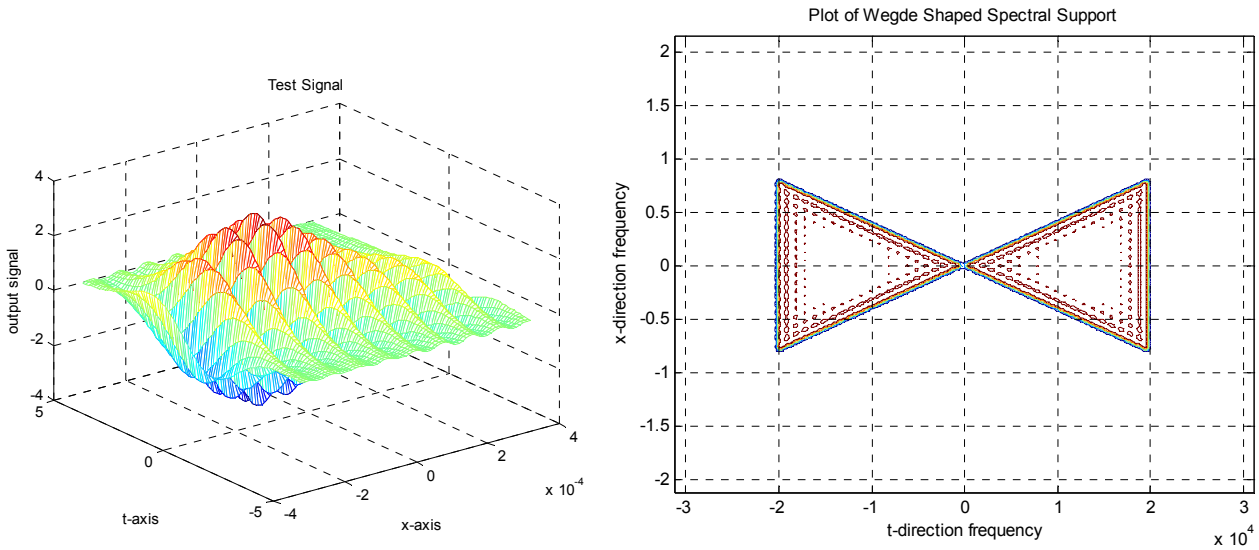

Fig. 8. Left: Reference test signal. Right: The spectral support region in 2-D frequency used in the Optimal Recovery interpolation.

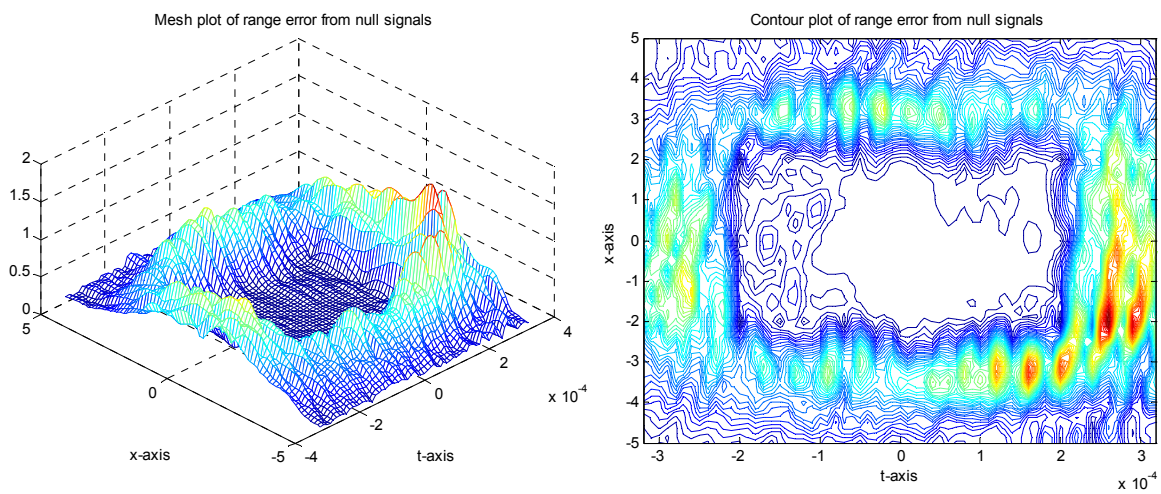

Fig. 9. Left: Mesh plot of maximum values of null signals (range errors) in a region surrounding and including the region of interest where all sensors are located. Right: Contour plot of range error.

3.6 Random Deployment of Additional Sensors for 2-D signals

Random deployment of additional sensors is mainly implemented in order to compare with optimal deployment. The same signals and sensor locations are used as in the previous subsection. Here instead of finding maximum range error location to place an additional sensor, another randomly located sensor is introduced at every iteration. The simulation is performed for 10 iterations. The metrics used for comparison of optimal vs. random deployment of sensors are the maximum and the average range error over the entire 2-D ROI where the existing and the new sensors are located.

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Figure 10 (left) shows the plot of maximum range error values of the optimally and randomly deployed additional sensors. Clearly, the curve for optimal deployment decreases much faster than that for random deployment. The value of maximum range error within the ROI for the 10th additional sensor is nearly equal to 0.2 for optimal case whereas it is nearly 0.98 for the random case. The plots of average range error for the optimal vs. random deployment cases are shown in Fig.10 (right). Again, we see that optimal deployment reduces the average range error by a larger amount.

The location of additional sensors for both the optimal and the random deployment methods are shown in Fig.11. In these plots, the blue color stars which are surrounded by red color circles are the 99 random samples which are used initially for both cases. The red color circles determine the new sample/sensor locations introduced by each method.

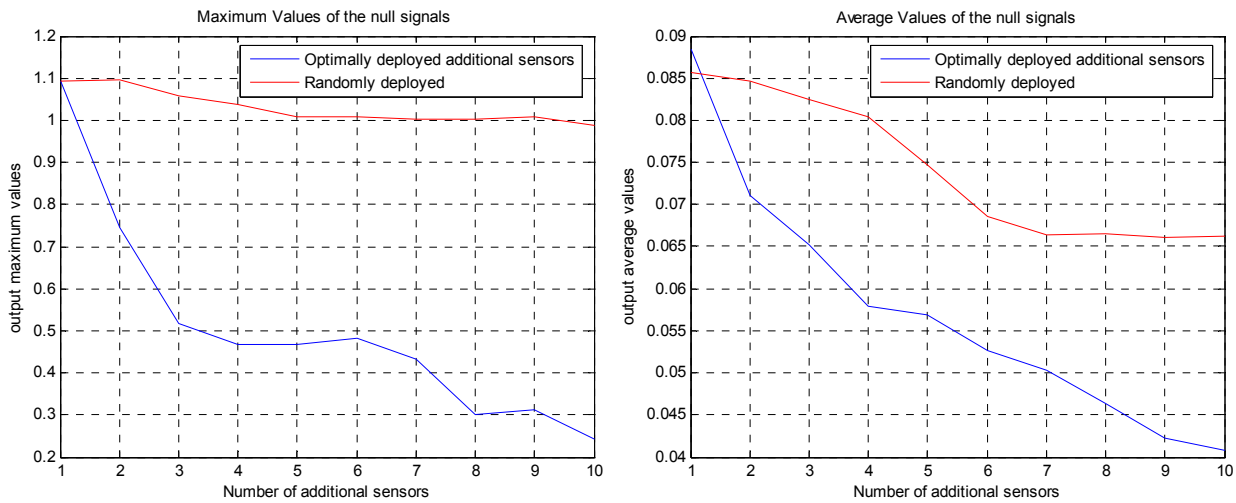

Fig. 10. Left: The maximum range errors in the ROI for optimally and randomly deployed additional sensors. Right: The average range error in the ROI.

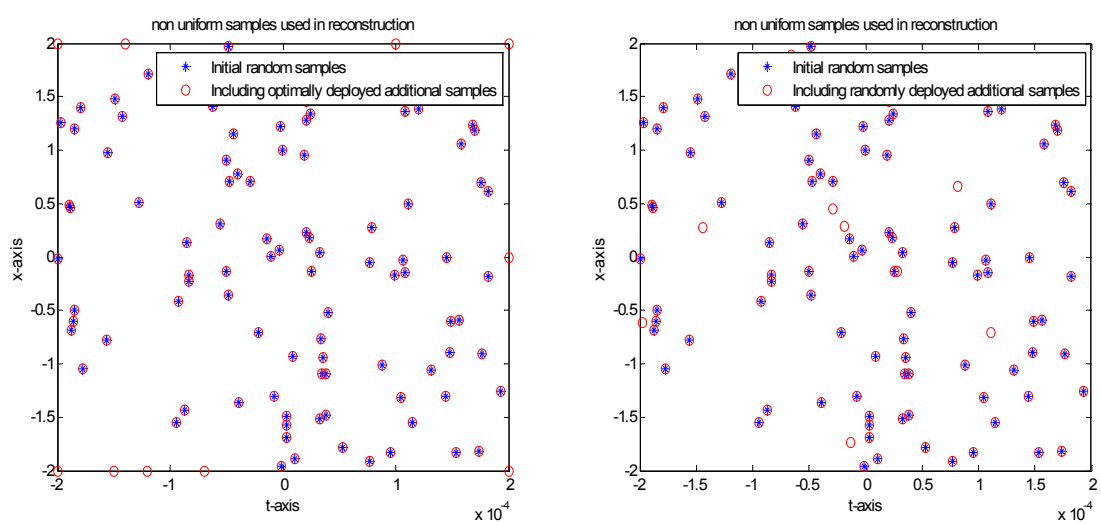

Fig. 11. Left: Optimally deployed additional sensors using optimal recovery framework. Right: Randomly deployed additional sensors not using optimal recovery framework

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4. CONCLUSIONS

In this paper we reviewed the theory of optimal recovery based interpolation including the error bounds for the estimation of a single arbitrary sample and for multiple samples. We established and illustrated an approach to generate feasible signals, null-space signals, extremal signals, and error bounds by computer simulation in order to more easily evaluate the problem of interpolation within a selected region of interest. Next we established a method for sequential addition of sensors based on placing new sensors at the location of the maximum error bound within the selected region. We illustrated the superiority of this approach over the alternative of random placement of additional sensors for 1-D and 2-D bandlimited signals.

5. ACKNOWLEDGEMENTS

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